TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders

Words: at least ≥ · at most ≤ · more than > · fewer than <.

Solve like an equation: undo + and −, then × and ÷.

Flip the sign when you multiply or divide both sides by a negative number.

Graph: ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

1. What is the solution set for $-3x + 4 \geq 19$?

A. $x \geq -5$

B. $x \leq 5$

C. $x \geq 5$

D. $\boldsymbol{x \leq -5}$

$-3x \geq 15 \rightarrow$ divide by −3, flip $\rightarrow x \leq -5$

2. Solve the inequality, and graph the solution on the number line.

$5 - 2x < 11$

$x > -3$, open circle at -3, shaded right

$-2x < 6 \rightarrow x > -3$ (flip)

3. A tire has 26 psi and loses 0.1 psi per minute. It must stay at or above 22 psi. Write and solve an inequality for the time t in minutes.

$t \leq 40$ minutes

$26 - 0.1t \geq 22 \rightarrow -0.1t \geq -4 \rightarrow t \leq 40$